

DEEPAK ARAVINDAN

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PROFESSIONAL SUMMARY

AI Product Engineer with 6+ years of combined experience building and shipping AI-powered products end-to-end — from defining product vision to writing production code. Hands-on with LLM orchestration, AI agent architectures, RAG pipelines, computer vision, and generative AI, backed by deep full-stack engineering skills (Python, React/Node.js, AWS, Docker). Proven ability to prototype rapidly, ship AI features that drive measurable user outcomes (30–40% faster deployments, 15–22% adoption lifts), and collaborate across ML engineering, design, and business teams. Author of a production, open-source video management system (AGPL-3.0) deployed at industrial sites.

TECHNICAL SKILLS

AI & ML:	LLM Orchestration (GPT-4, Claude, LLaMA, Mistral), AI Agents & Tool Use, RAG Architecture, Prompt Engineering & Optimization, Fine-Tuning (LoRA), Computer Vision (YOLO, OpenVINO, OpenCV), NLP, Model Evaluation, Hallucination Reduction, Guardrails & Safety
Languages & Frameworks:	Python, JavaScript/TypeScript, React, Node.js, Express, FastAPI, REST APIs, GraphQL, Kafka, SQL
Infrastructure & MLOps:	AWS (Lambda, EC2, S3, SageMaker), Docker, CI/CD Pipelines, FFmpeg, GStreamer, MediaMTX, ONVIF, RTSP/HLS, Keycloak (OIDC), Microservices, Edge AI Deployment, Model Serving
Data & AI Tooling:	LangChain, LlamaIndex, Vector Databases (pgvector, Weaviate, Pinecone), PostgreSQL, Valkey/Redis, OpenAI API, CLIP & Text Embeddings, Mixpanel, Google Analytics, A/B Testing, Data Pipelines
Product & Design:	PRDs, Roadmapping, User Stories, RICE/MoSCoW Prioritization, Figma, Miro, Agile/Scrum, User Research, Competitive Analysis
Tools:	Jira, Git, GitHub Actions, Postman, Jupyter Notebooks

PROFESSIONAL EXPERIENCE

AI Product Engineer — Variphi

Bangalore, India · Jan 2025 – Present

- Architected, shipped and open-sourced the **Intelligent Video Management System** ([GitHub](#), AGPL-3.0) — a self-hosted NVR running 24/7 across 9+ industrial cameras: ONVIF discovery, single-pull MediaMTX RTSP relay, codec-copy recording, and a shared frame broker that decodes each stream once for every AI consumer. Owned it end-to-end, from camera ingest to operator console.
- Ran YOLO via OpenVINO for person, vehicle, plate and face detection on **commodity CPUs with no GPU**, cutting hardware cost per site; forensic search over CLIP embeddings in pgvector turns weeks of footage into a plain-English query.
- Built the **Live Event Assistant**, a RAG agent answering natural-language questions over a live safety-event stream (6,800+ events, 24 locations). Cut hallucination on quantitative questions by moving **all counting out of the model** — retrieval returns database-computed aggregates and the LLM only phrases them and picks a verifiable sample event. Pluggable model layer: self-hosted open LLM when data cannot leave the customer's hardware, OpenAI API when it can.
- Architected and shipped a multi-agent AI system for real-time safety-violation monitoring, orchestrating LLM-based agents with tool-use capabilities (LangChain, OpenAI API) for autonomous image analysis and alert generation — reduced client deployment time by 30%.
- Built end-to-end generative AI pipeline for on-demand image synthesis, implementing prompt optimization, guardrails for content safety, and evaluation frameworks to ensure output quality at production scale.
- Designed and optimized HALO Accelerator — a zero-touch provisioning system (Python, Docker, AWS Lambda) — cutting enterprise onboarding from weeks to days (40% reduction) and enabling self-serve configuration for new clients.
- Led R&D evaluation of foundation models (GPT-4, Claude, open-source LLMs) across generative and perceptive AI use-cases; owned the product roadmap and technical PRDs for all AI features.

Product Engineer — ValueMatrix.ai

Bangalore, India · Feb 2023 – Dec 2024

- Owned end-to-end roadmap and hands-on development for a candidate evaluation platform used by enterprise clients — defined PRDs, wrote user stories, and personally shipped core features, improving client retention by 22%.
- Led a small engineering team to build LinkedIn integration for real-time candidate scoring and a Kafka-based analytics pipeline, reducing manual tasks by 60% and cutting API latency by 30%.
- Designed and launched Big5 psychometric assessment module end-to-end — from product spec to frontend implementation to release — driving 30% increase in upsell revenue.
- Established security best practices across the MERN stack: built middleware for input validation and XSS prevention, lowering vulnerabilities by 40% and improving platform reliability.
- Conducted user research and competitive analysis, informing two high-ROI feature releases; reduced feature delivery cycle by 18% and boosted UX task completion by 25%.
- Built Mixpanel & GA dashboards with custom event tracking; iterated UI based on data insights and ran A/B tests on activation flows, lifting activation rates by 19% within four weeks.

Frontend Engineer & IoT Developer — Autonomo Technologies

Bangalore, India · Apr 2021 – Jan 2023

- Architected and shipped React-based admin portal for autonomous store operations powered by computer vision checkout — reduced operational delays by 35%.
- Built "Hoody" — a real-time fault-tracking system for CV-based checkout hardware, integrating IoT sensor telemetry for predictive diagnostics; reduced support escalations by 40%.
- Integrated IoT peripherals (sensors, dispensers) with cloud backend, enabling automated device management and laying groundwork for AI-driven predictive maintenance.

Software Engineer (Contract) — Smartlink / DIGISOL

Bangalore, India · 2020 – 2021

- Developed scalable router management platform handling 50K+ devices with 99.9% uptime; implemented security hardening and performance optimizations for production infrastructure.

OPEN SOURCE & AI PROJECTS

- **Intelligent Video Management System** — [GitHub](#) (AGPL-3.0). Self-hosted VMS: FastAPI microservices, React console, MediaMTX relay, YOLO/OpenVINO analytics, CLIP + pgvector forensic search, Keycloak RBAC across 5 tiers, Docker Compose deploy. Public code, in production at an industrial site.
- **RAG Knowledge Assistant**: internal documentation search using LangChain, Weaviate, and OpenAI embeddings — reduced information retrieval time by 50% across the engineering team.
- **AI Resume Screener**: end-to-end candidate-job matching prototype using fine-tuned transformer models and prompt engineering — 92% accuracy on benchmark dataset.
- **Multi-Agent Task Automation**: autonomous agent workflow using LangChain agents with tool-use (web search, code execution, API calls) for automated research and reporting.
- **MLOps Pipeline**: model versioning, automated evaluation, and CI/CD deployment on AWS SageMaker for continuous model improvement and A/B model serving.

CERTIFICATIONS

- Certified Scrum Product Owner (CSPO)
- Generative AI with Large Language Models — DeepLearning.AI & AWS
- Google Analytics Individual Qualification (GAIQ)
- Google UX/UI Design Professional Certificate

EDUCATION

- B.Tech, Computer Science — SRM University